

Final Design Report & Data Analysis

EDX Silicon-Photonix Design Course
(by *Lukas Chrostowski*)

Reporter Name: *Faingersh Alexander*

Course User: *fainalexander*

Fork Name: [fainsasha](#) / [openEBL-2025-10](#)

Email: *alexfa@towersemi.com*

General Introduction

Introduction to Photonics

Photonics is the science and technology of generating, controlling, and detecting light. Similar to how electronics relies on the flow of electrons, photonics makes use of photons to perform a wide range of functions. Because light can carry information at very high speeds and with low energy loss, photonics has become a key enabler of modern technologies

Integrated Silicon Photonics

Integrated silicon photonics allows the implementation of compact optical circuits that use light to perform tasks similar to those in electronics, but at much higher speeds. These circuits can handle functions such as computing, data transfer, and signal modulation. Applications span across telecommunications, data centers, quantum computing, biomedical diagnostics, and scientific research, among many others.

A major advantage of integrated photonics is its compatibility with established semiconductor manufacturing processes. Many of the same concepts used in integrated electrical circuits apply, and standard CMOS foundry technology—even older process nodes—can be repurposed for fabricating photonic integrated circuits (PICs)

Focus of This Design

This project focuses on the design of an **Imbalanced Mach–Zehnder Interferometer (IMZI)**, an optical device that can be used to extract key parameters of silicon strip waveguides, such as the Free Spectral Range (FSR) and the group index

Imbalanced Mach-Zender Interferometer (IMZI)

The **Mach–Zehnder Interferometer (MZI)** is one of the most widely used and versatile devices in integrated photonics. At a higher level, MZIs appear in a broad range of applications:

- **Telecommunications & Data Centers** – as high-speed optical modulators, wavelength filters in WDM systems, and optical switches
- **Sensing & Metrology** – for refractive-index detection, biosensing, gas and temperature sensing, and general phase-sensitive measurements
- **Quantum Photonics** – serving as beam splitters, interferometric quantum gates, and tools for generating and manipulating entanglement
- **Photonic Computing & Signal Processing** – in photonic neural networks, matrix multipliers, and microwave photonic filters
- **Research & Characterization** – for extracting waveguide parameters (effective index, group index, loss) and validating fabrication processes.

This wide applicability comes from the MZI's ability to precisely control light interference. When the two optical paths have different lengths, the device is referred to as an **Imbalanced MZI (IMZI)**

An IMZI introduces a **relative phase shift** between two light beams traveling along waveguides of different lengths. The path length difference is expressed as: $\Delta L = L_1 - L_2$. It causes phase delay between branches and determines the resulting output intensity

IMZI basic

An IMZI introduces a **relative phase shift** between two light beams traveling along waveguides of different lengths. The path length difference is expressed as: $\Delta L = L_1 - L_2$. It causes phase delay between branches and determines the resulting output intensity

$n_{eff} \equiv$ **effective index**

$\beta \equiv$ **propagation constant**

$n_g \equiv$ **group index** of the waveguide

$\Delta L \equiv$ **Length difference** ($L_1 - L_2$)

$\Delta\phi \equiv$ **phase delay** (due to ΔL f.ex)

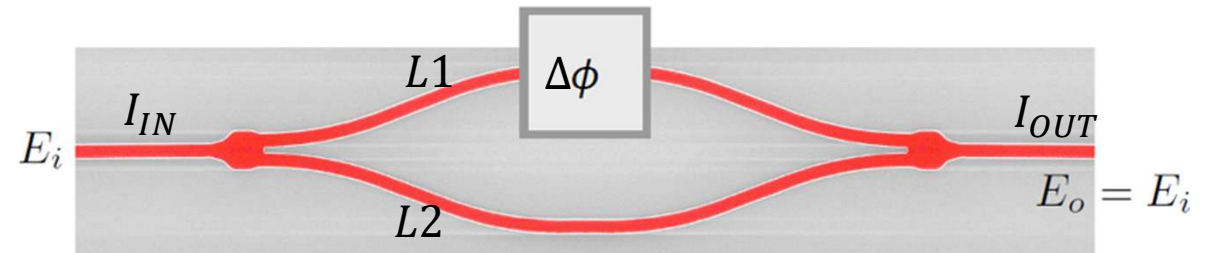
$$\beta = \frac{2\pi}{\lambda} \left[\frac{rad}{m} \right]$$

$$\beta_{eff} = \beta n_{eff}$$

$$\Delta\phi = \beta_{eff} \Delta L$$

$$I_{OUT} = \frac{I_{IN}}{2} (1 + \cos \Delta\phi) = \frac{I_{IN}}{2} \left[1 + \cos \left(\frac{2\pi}{\lambda} n_{eff} \Delta L \right) \right]$$

Interferometer built using Y-Branches



The **Free Spectral Range (FSR)** is the spacing between two neighboring peaks (or dips) in the transmission spectrum of an interferometer.

FSR actually tells how far apart the resonance peaks are in wavelength

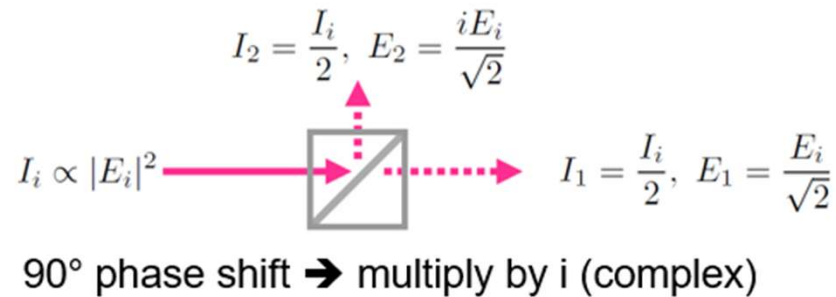
$$FSR_{(\Delta L)} = \frac{\lambda^2}{n_g \Delta L}$$

$$n_{g(\Delta L)} = \frac{\lambda^2}{FSR * \Delta L}$$

Measure **FSR [m]** → Calculate **n_g** → Compare to the theoretical

Beam Splitter/Combiner basic

I – Intensity
E – Electric field



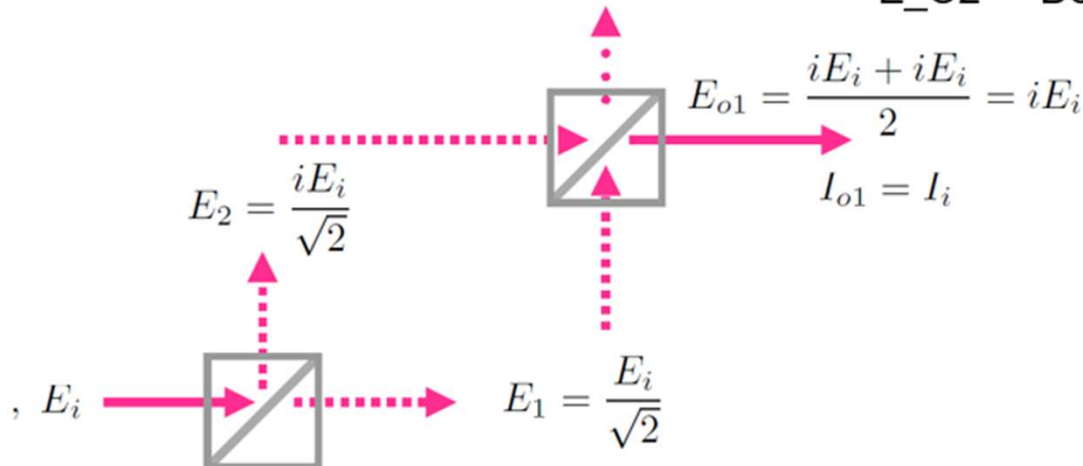
Interferometer using beam splitter and combiner

• Interferometer:

$$E_{o2} = \frac{i^2 E_i + E_i}{2} = 0$$

E_O1 - Same Input Intensity, 90° phase shift

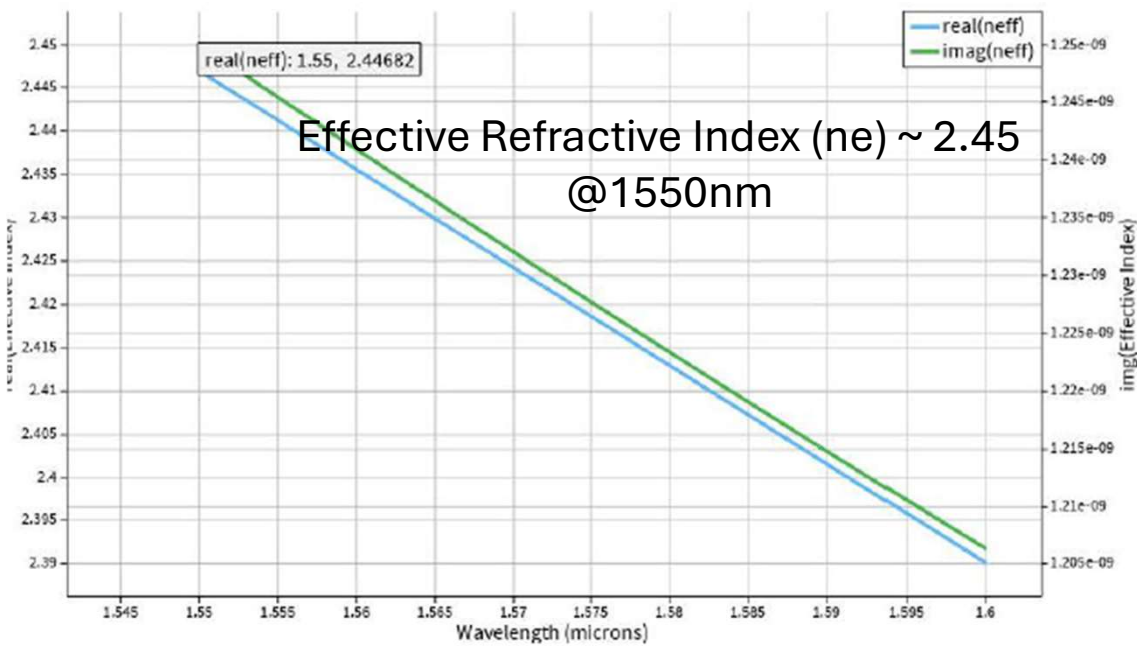
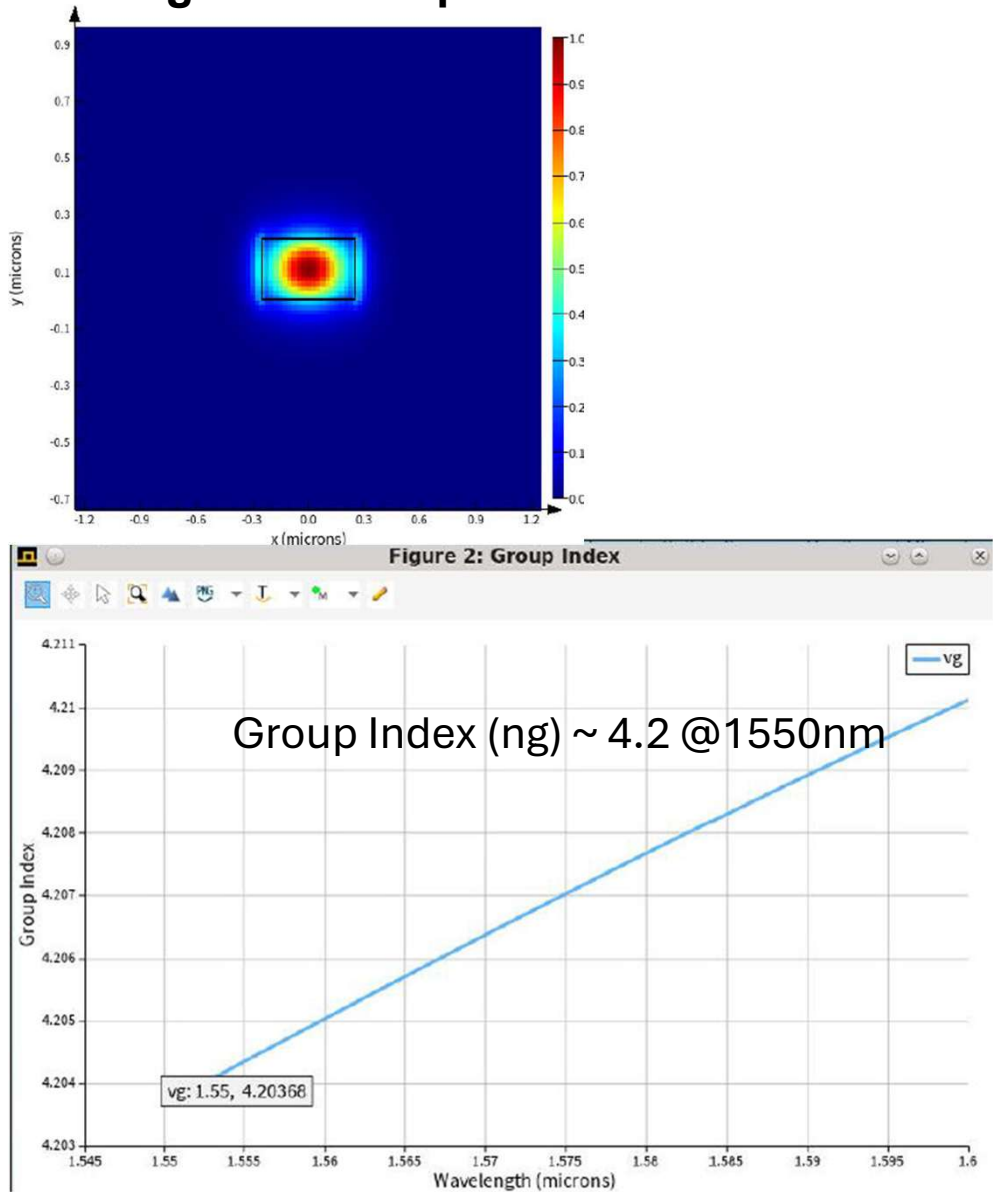
E_O2 - Destructive interference



$$E_{o1} = \left[\frac{i \left(\frac{E_i}{\sqrt{2}} \right)}{\sqrt{2}} + \frac{\left(\frac{iE_i}{\sqrt{2}} \right)}{\sqrt{2}} \right] = iE_i$$

$$E_{o2} = \left[\frac{i \left(\frac{iE_i}{\sqrt{2}} \right)}{\sqrt{2}} + \frac{\left(\frac{E_i}{\sqrt{2}} \right)}{\sqrt{2}} \right] = 0$$

Waive-guide basic parameters/simulation

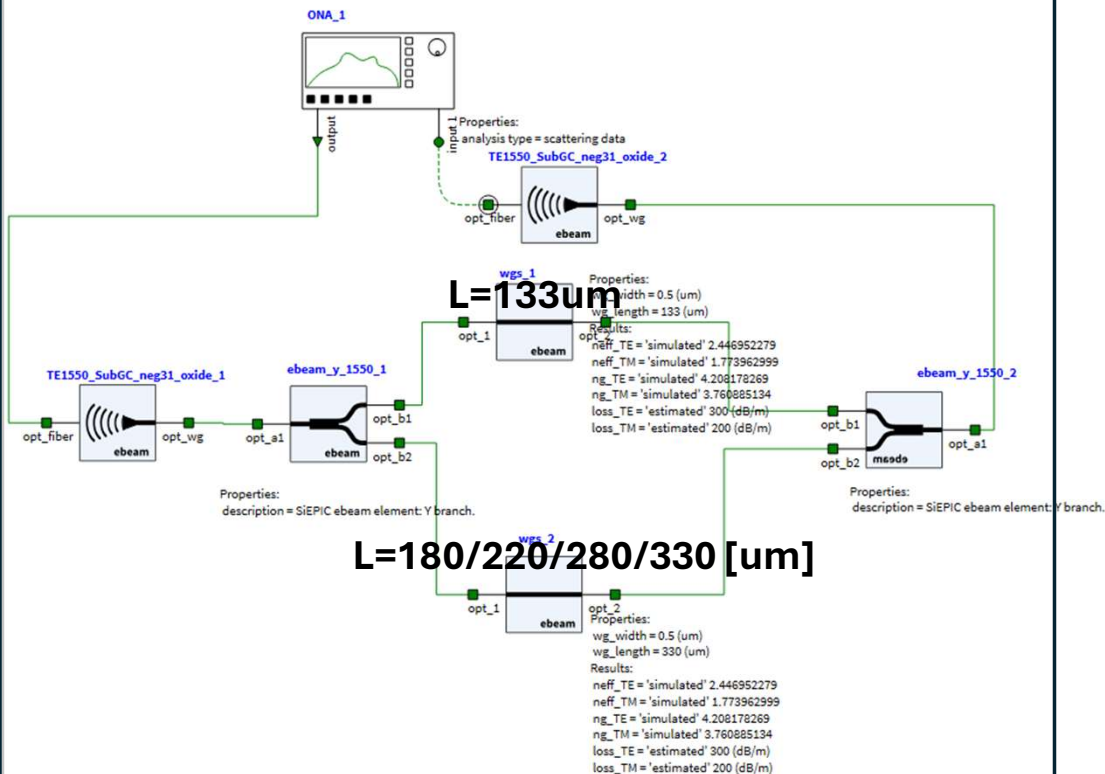


The waveguide is TE polarized for 1550nm wavelength. The wave guide mode profile (E-intensity) corresponding to TE mode, effective index and group index are illustrated in the following snapshots:

Name	Value
neff_TE	'simulated' 2.446952279
neff_TM	'simulated' 1.773962999
ng_TE	'simulated' 4.208178269
ng_TM	'simulated' 3.760885134
loss_TE	'estimated' 300 (dB/m)
loss_TM	'estimated' 200 (dB/m)

Pre-Layout simulation test-bench

4 x MZI based on 2 x Y-branches



Layout - unit

ebeam_gc_te_1550

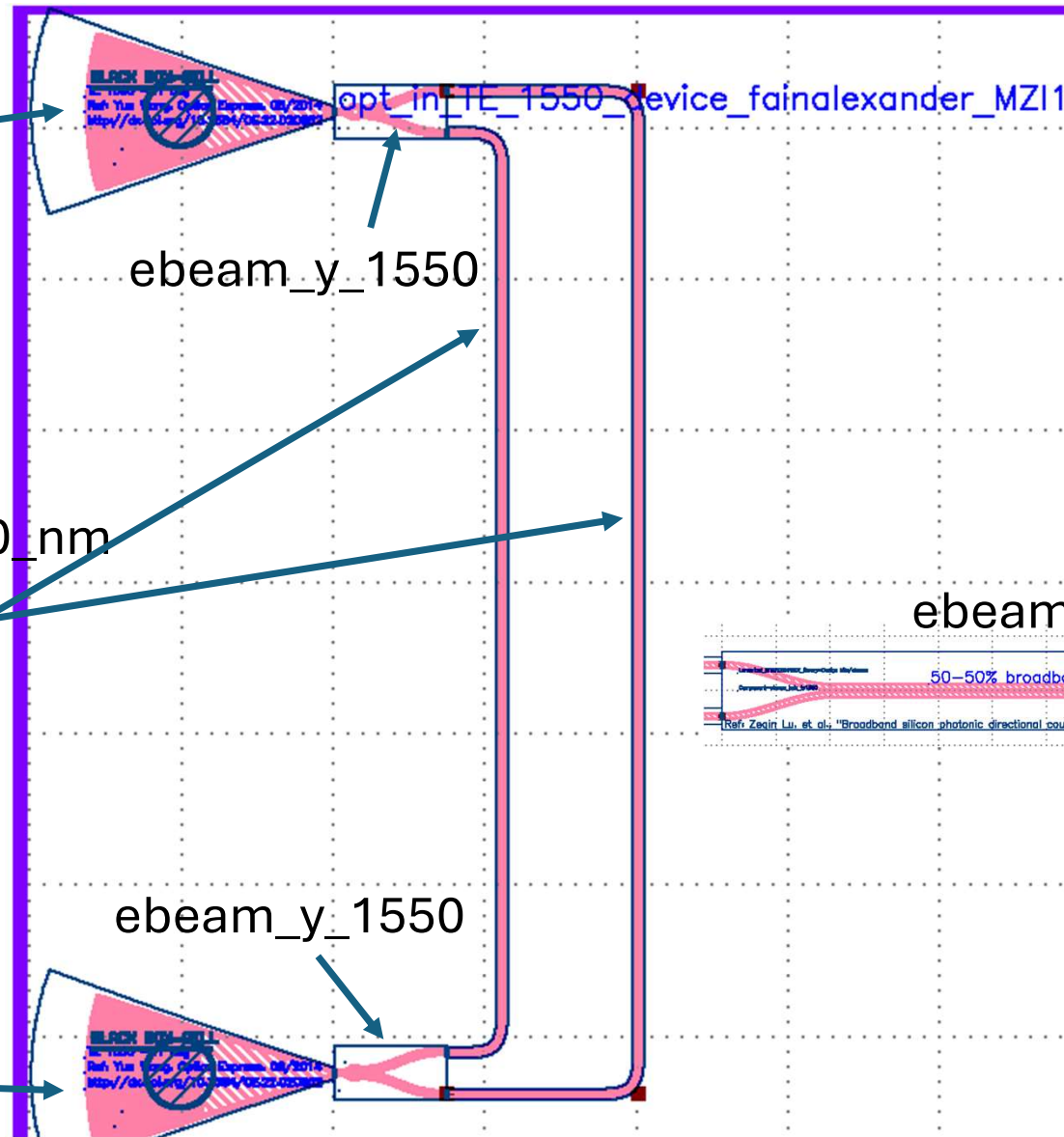
ebeam_y_1550

Waiveguide_Strip_TE_1550_nm
(w=500nm)

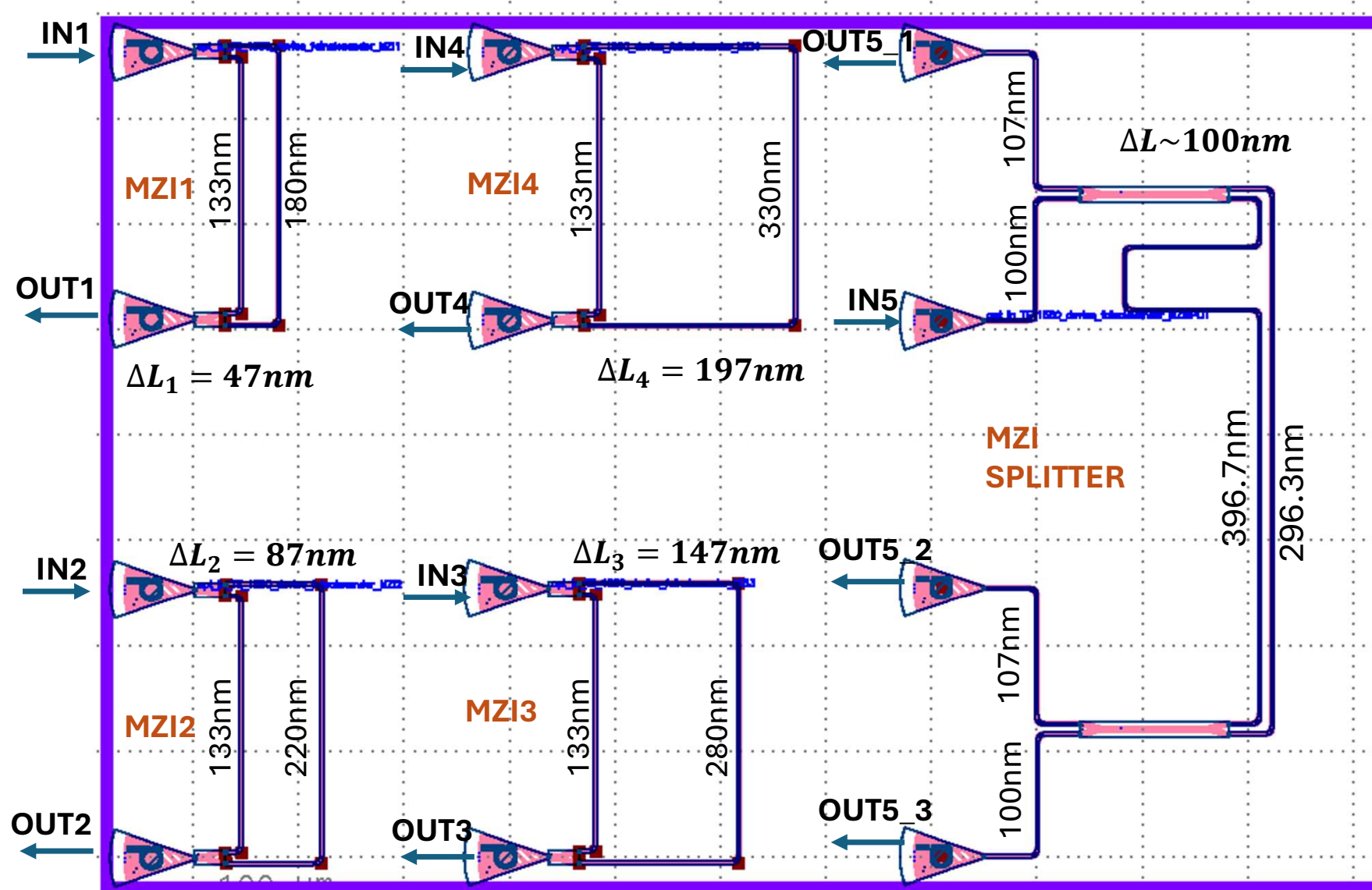
ebeam_bdc_te_1550

ebeam_y_1550

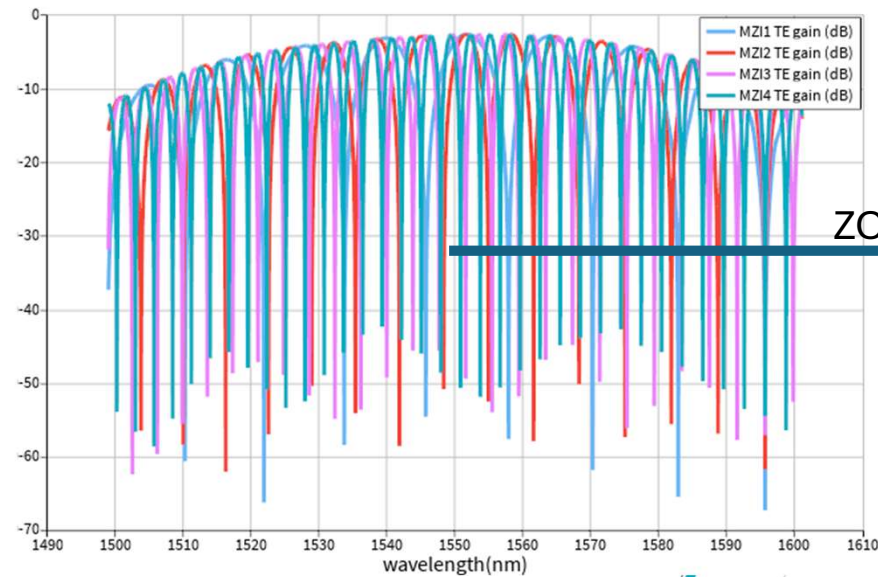
ebeam_gc_te1550



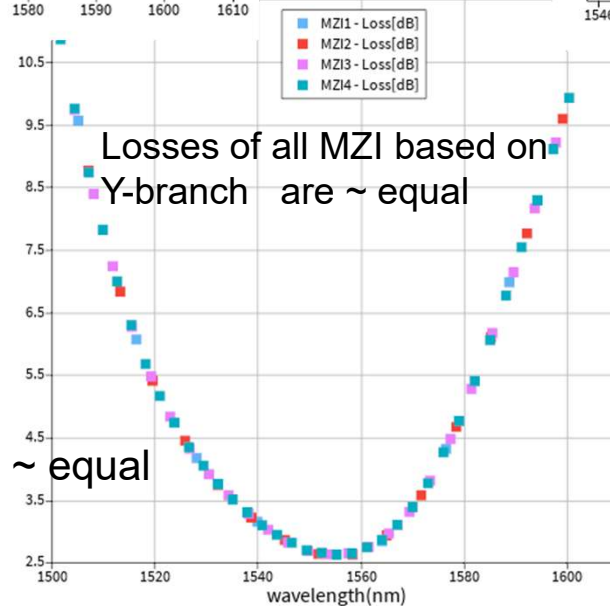
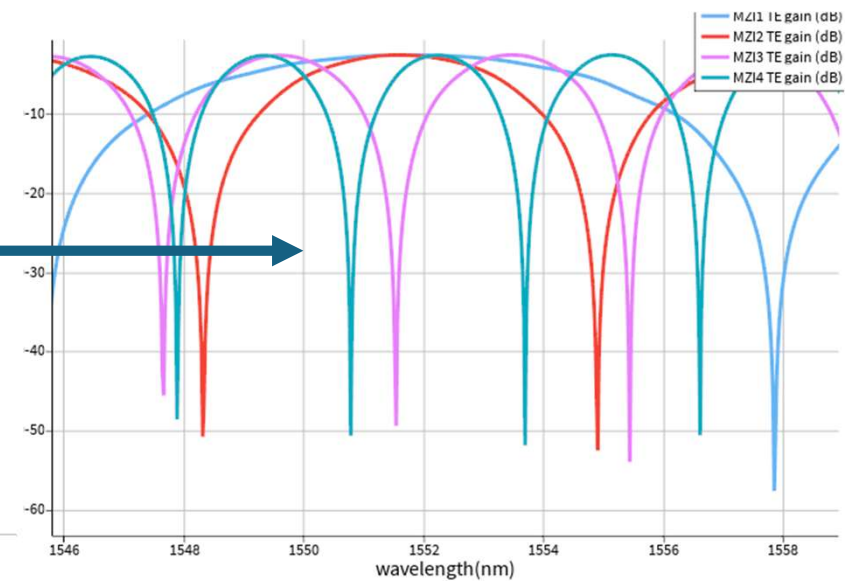
Layout – Full, MZI x 5



4 x Y-branch based MZIs pre-LO



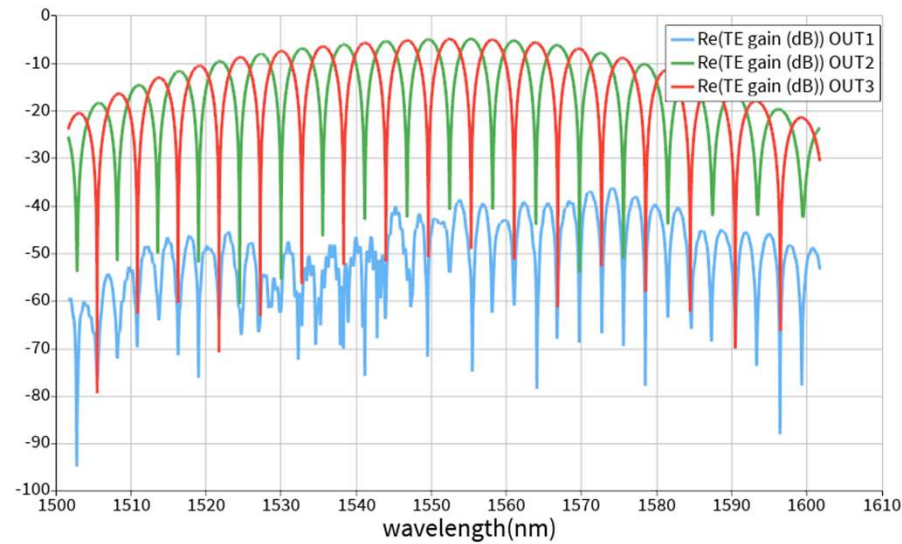
ZOOM on 1550nm



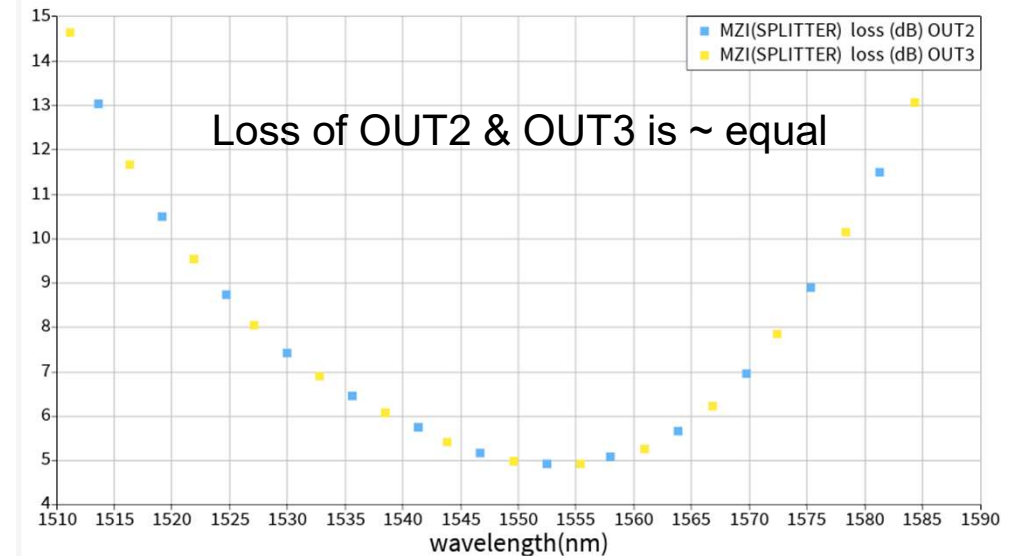
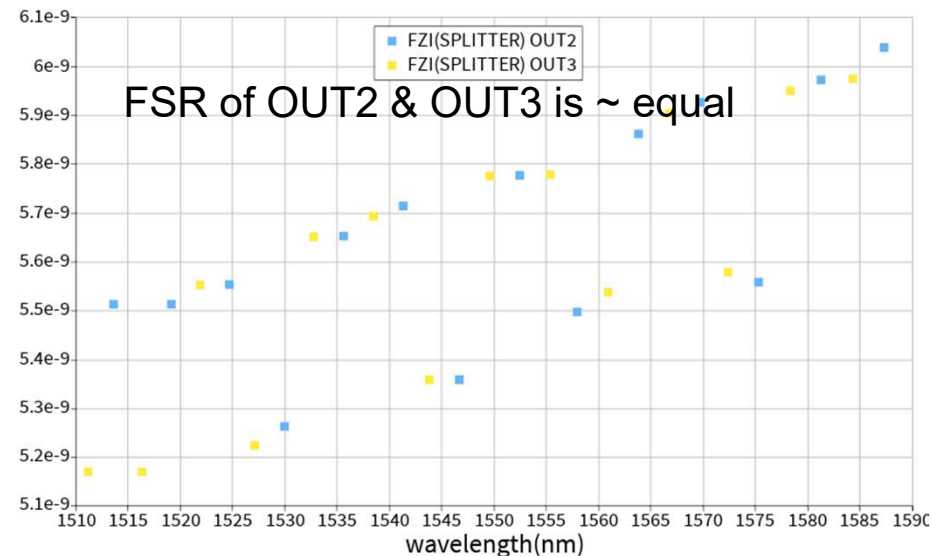
Losses of all MZI based on Y-branch are ~ equal

FSR of OUT2 & OUT3 is ~ equal

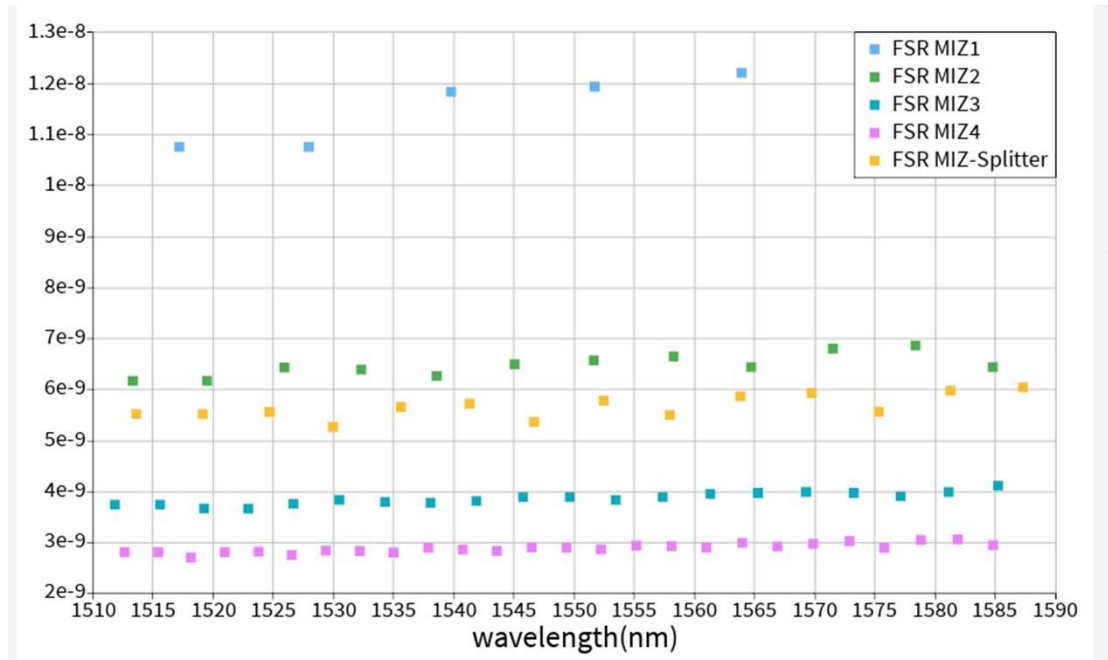
SPLITTER based MZI pre-LO



OUT2 & OUT3 are in anti-phase
And OUT1 is negligible as expected



FSR simulation results pre-LO. All 5 x MZI

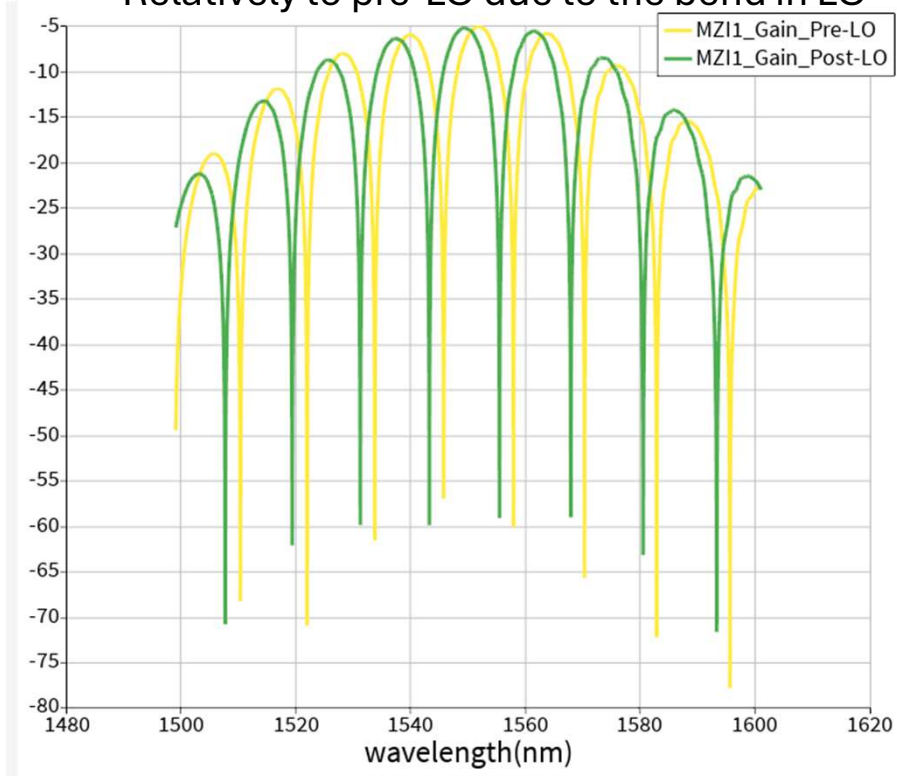


	$\lambda[\text{nm}]$			
	1.55E+03			
			$n_{G(\Delta L)} = \frac{\lambda^2}{FSR * \Delta L}$	
Device	L1[um]	L2[um]	FSR[nm]	ng
MZI1	133.00	180.00	11.90	4.30
MZI2	133.00	220.00	6.57	4.20
MZI3	133.00	280.00	3.89	4.20
MZI4	133.00	330.00	2.89	4.22
MZI_SPLITTER	296.30	396.70	5.60	4.27

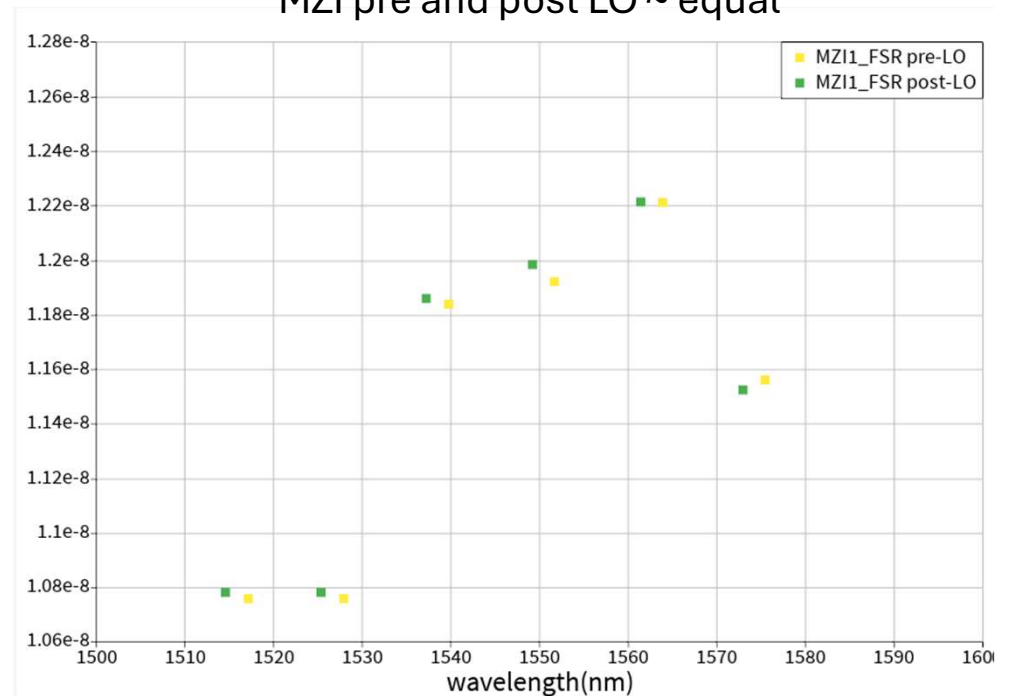
Group index calculated from FSR measurement fits the expected ~ 4.2

Pre LO Vs Post LO. MZI1 compared

Phase of post LO has a little shift
Relatively to pre-LO due to the bend in LO

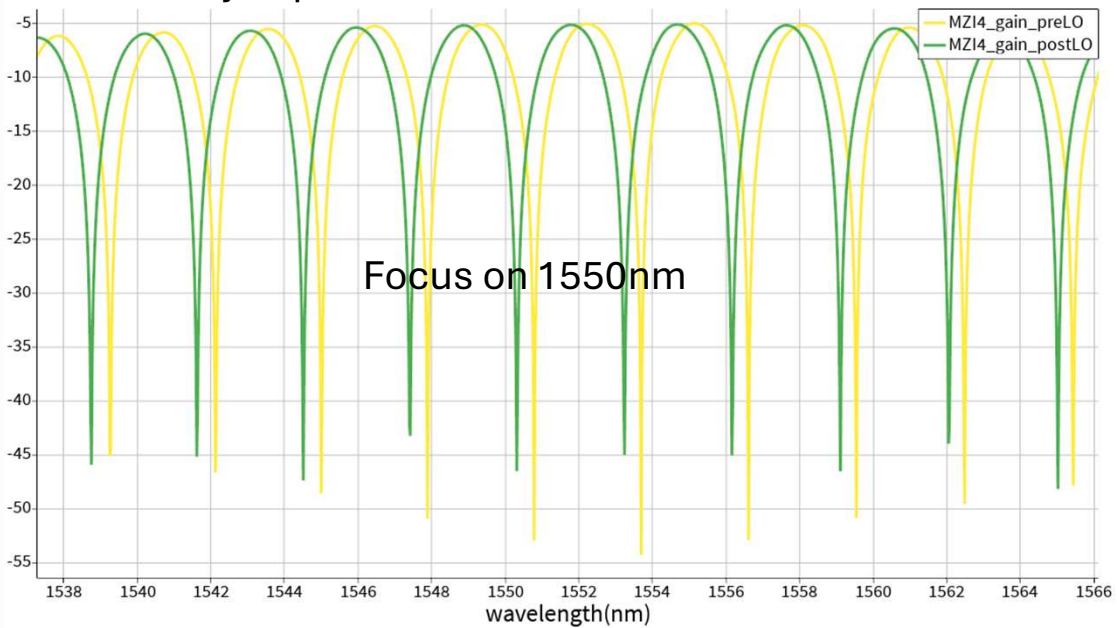


MZI pre and post LO ~ equal

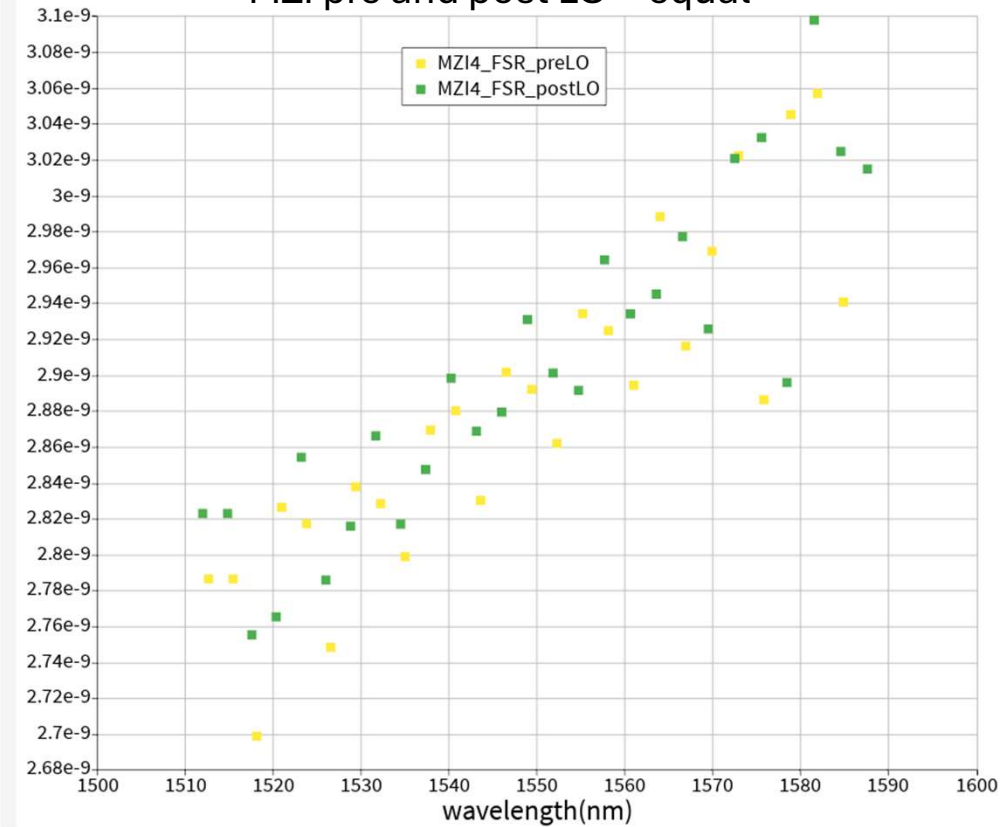


Pre LO Vs Post LO. MZI4 compared

phase of post LO has a little shift
relatively to pre-LO due to the bend in LO

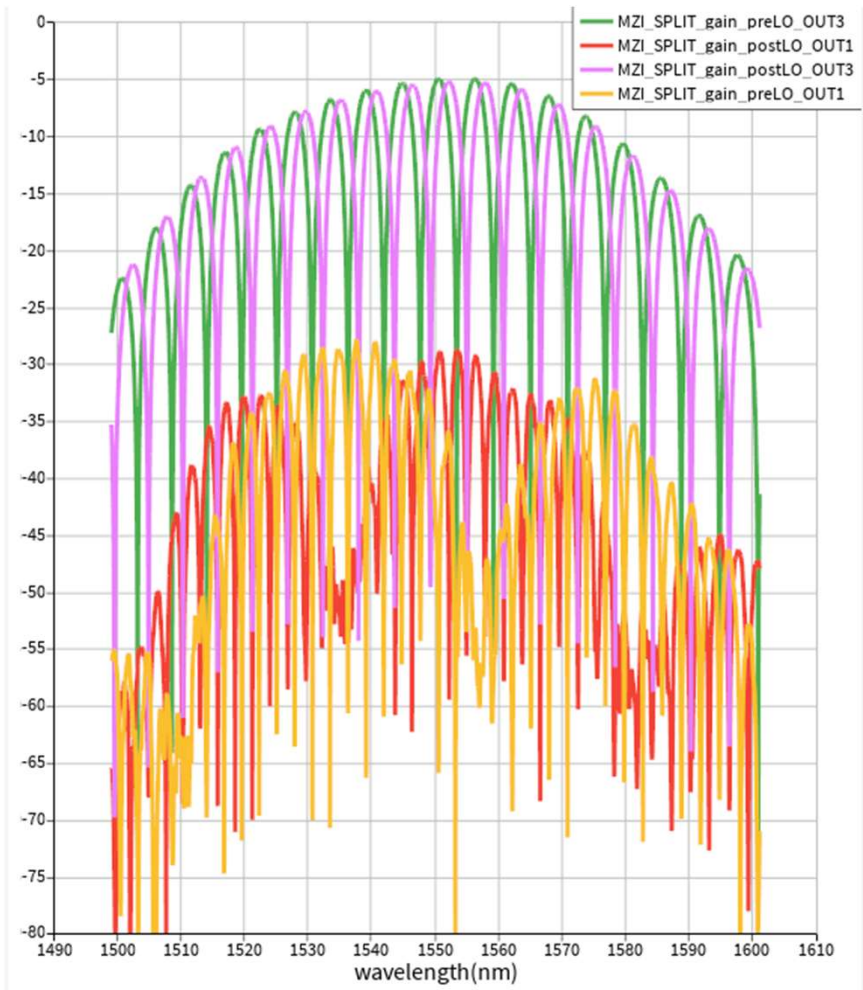


MZI pre and post LO ~ equal

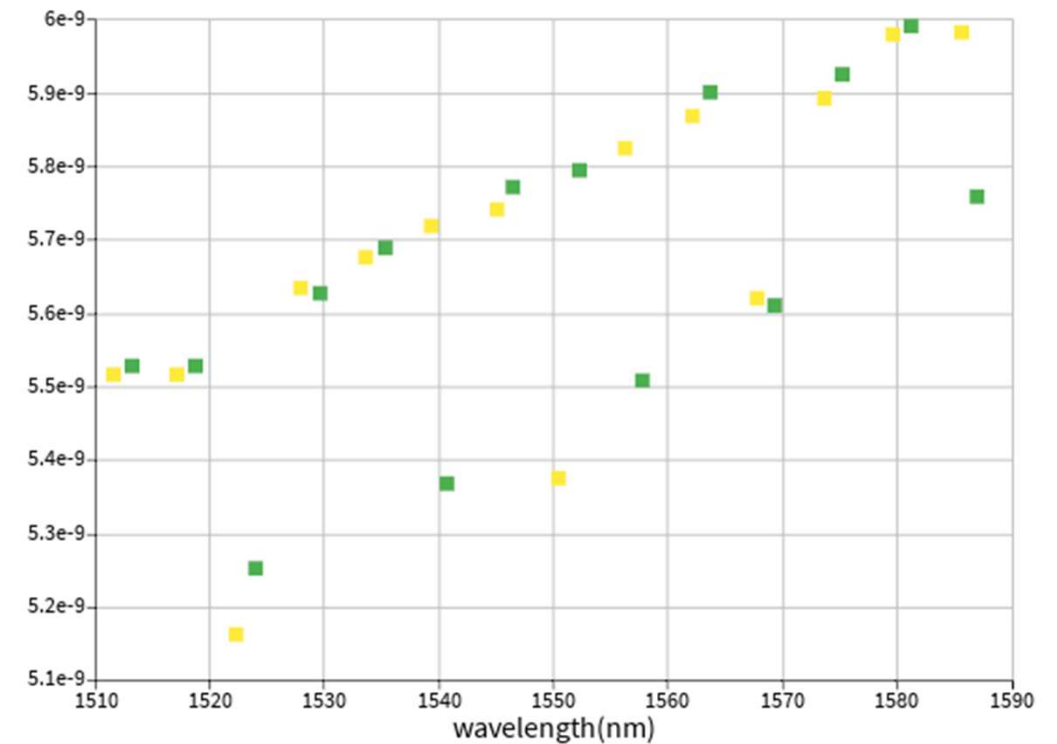


Pre LO Vs Post LO. MZI SPLITTER compared

phase of post LO has a little shift
relatively to pre-LO due to the bend in LO



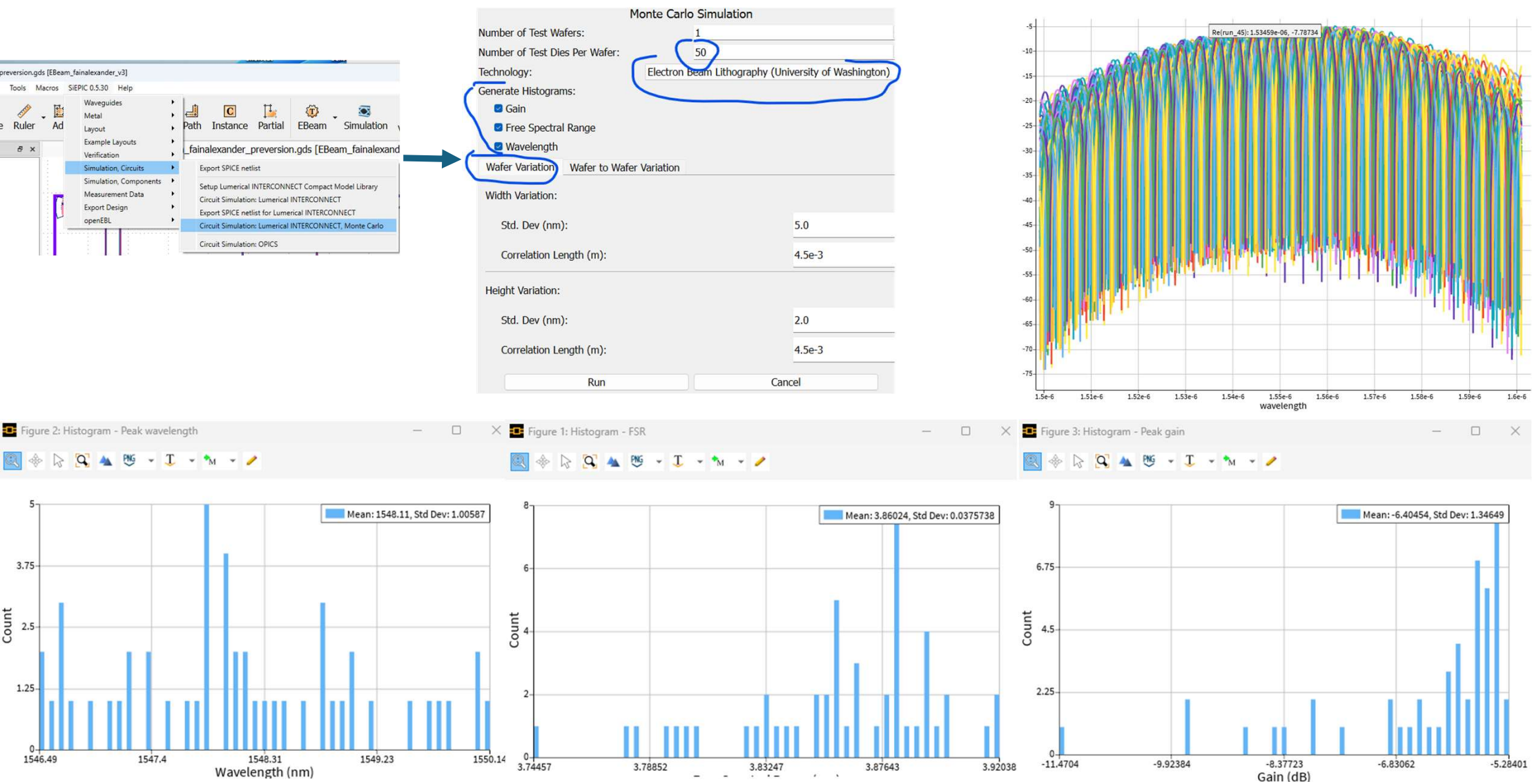
MZI pre and post LO ~ equal



Process (Width/Thickness) variations

	λ [nm]	ΔL [nm]	$FSR = \frac{\lambda^2}{n_G \Delta L}$		
	1.55E+03	1.00E+05			
Waive Guide Process	Width [nm]	Thickness [nm]	n_eff (simulated)	n_group (simulated)	FSR [nm] (calculated @ $\Delta L=100\mu m$)
Typical	500	220	2.45	4.204	5.71E+00
Process 1	470	215.3	2.372	4.25	5.65E+00
Process 2	470	223.1	2.404	4.262	5.64E+00
Process 3	510	215.3	2.442	4.18	5.75E+00
Process 4	510	223.1	2.472	4.19	5.73E+00

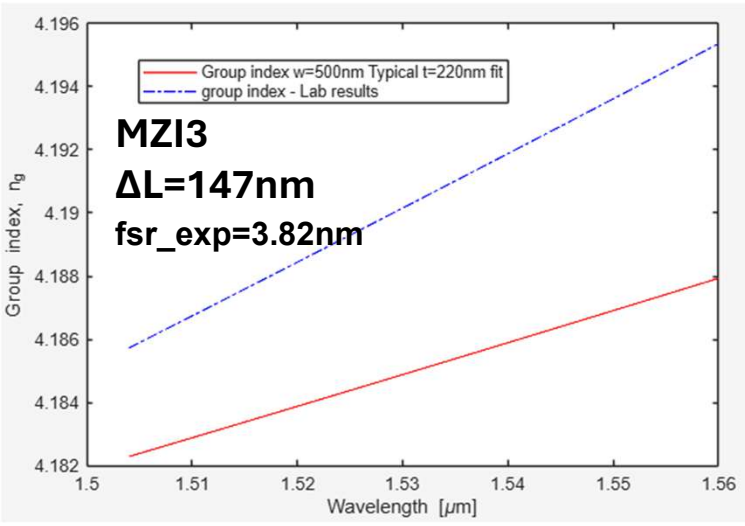
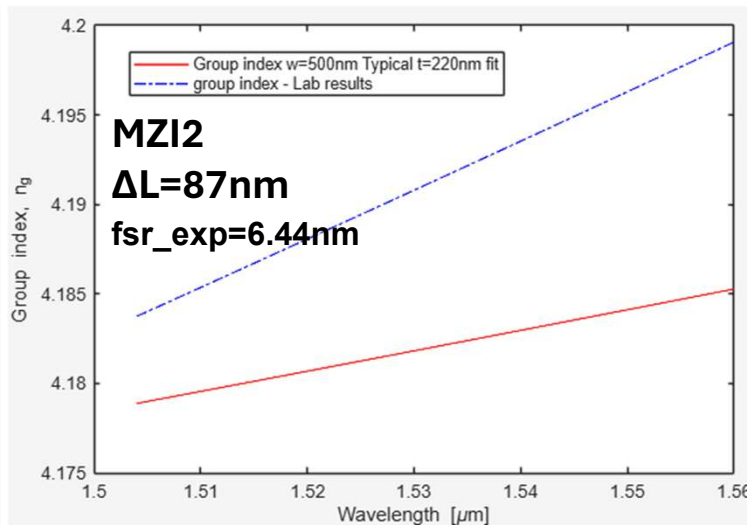
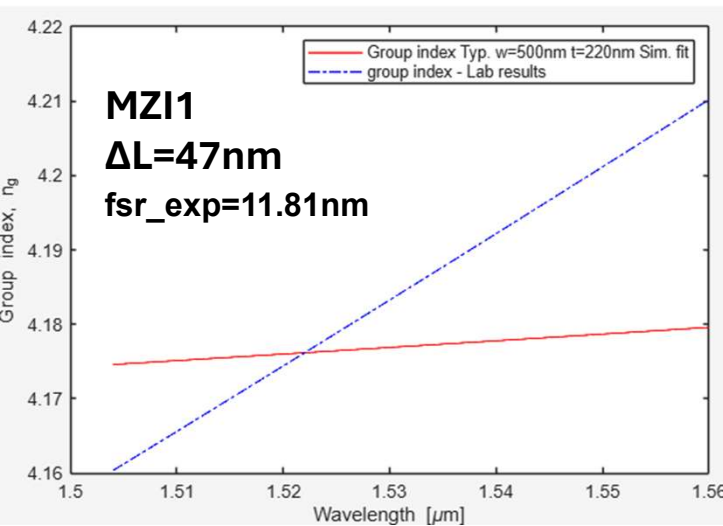
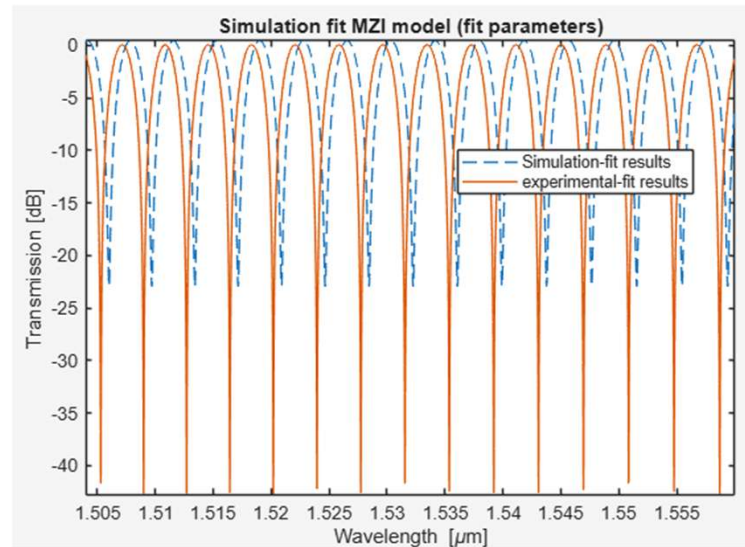
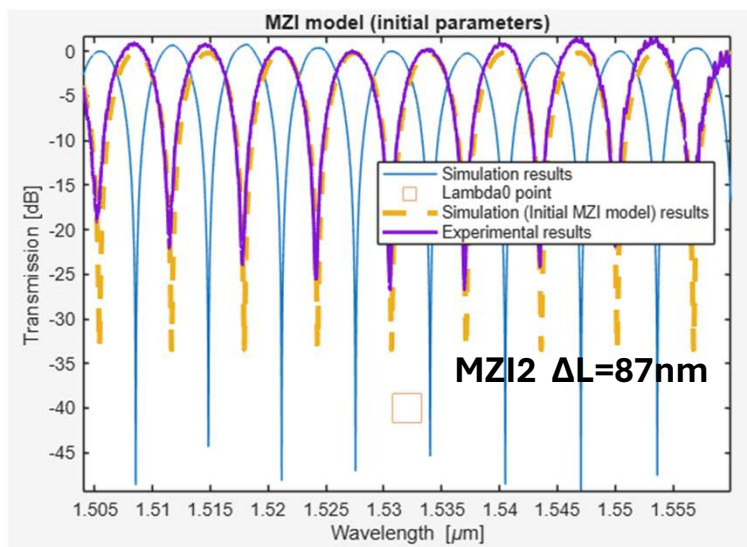
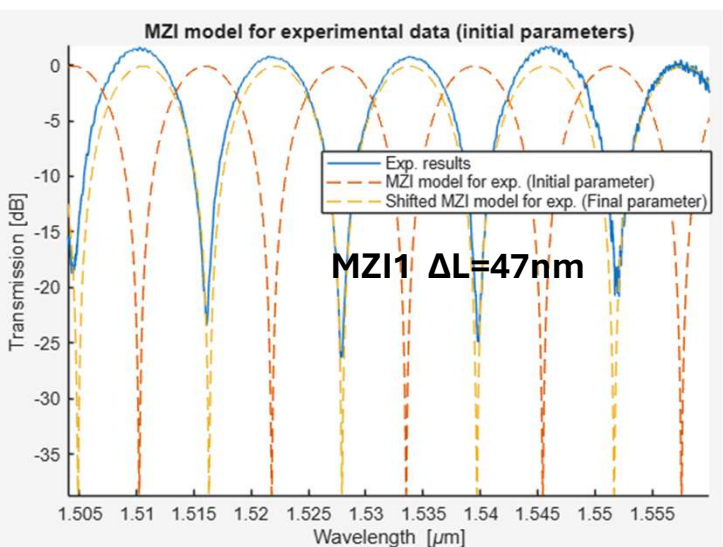
Monte – Carlo MZI3 example. Variations (on single wafer)



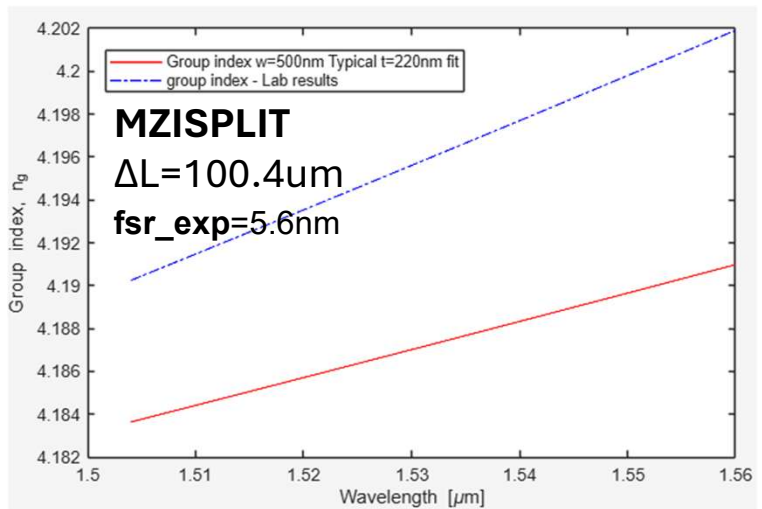
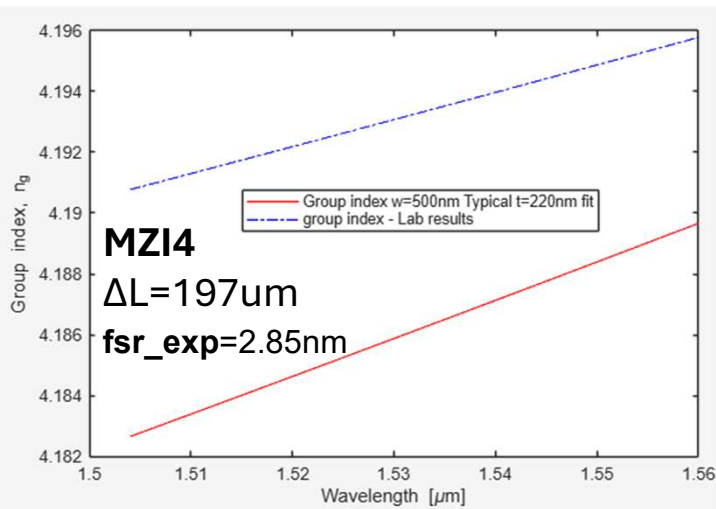
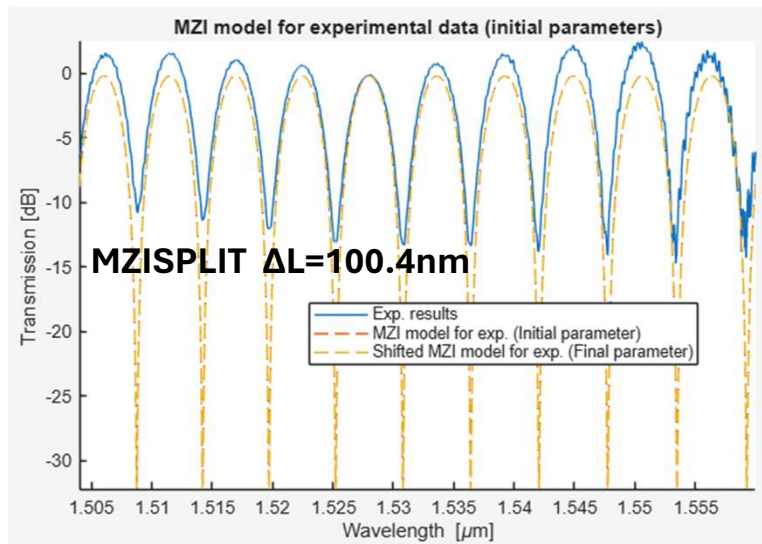
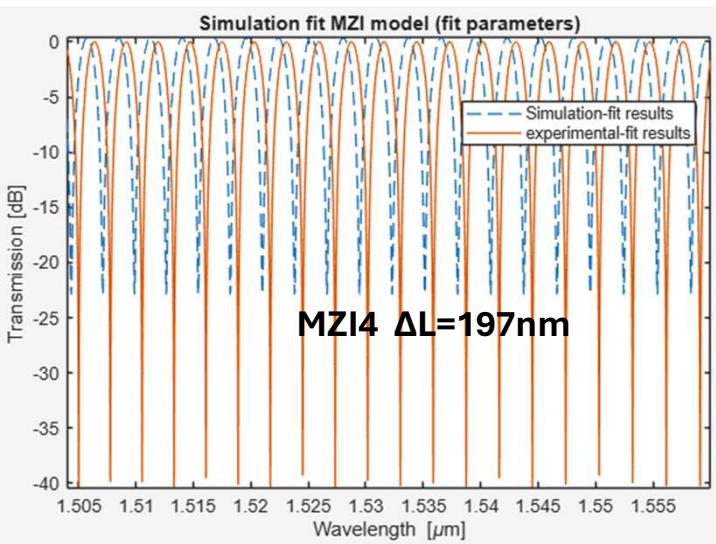
Monte – Carlo: Summary of all 5 x MZI Variations (on single wafer)

	Peak [nm]		Gain [dB]		FSR [nm]	
Monte Carlo	Mean	Sigma	Mean	Sigma	Mean	Sigma
MZI1	1548.7	3.02	-6.5	1.2	12.04	0.29
MZI2	1548.1	2.26	-6.9	3	6.66	0.16
MZI3	1548	1	-6.45	1.35	3.83	0.043
MZI4	1548.5	0.85	-6.4	1.38	2.89	0.089
MZISPLIT	1548.1	1.1	-9.8	3.4	3.35	1.5

Lab results Vs. typical simulation fit. *Note: Lab Laser @ 1532nm, Sim @ 1550nm*

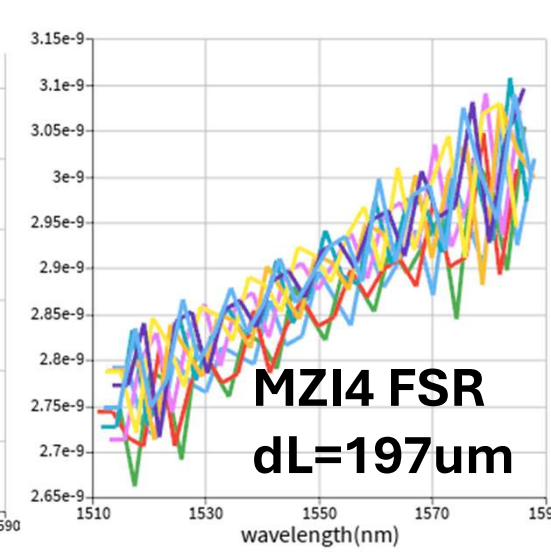
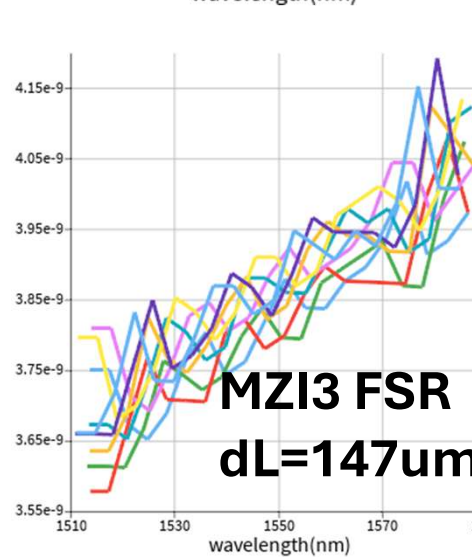
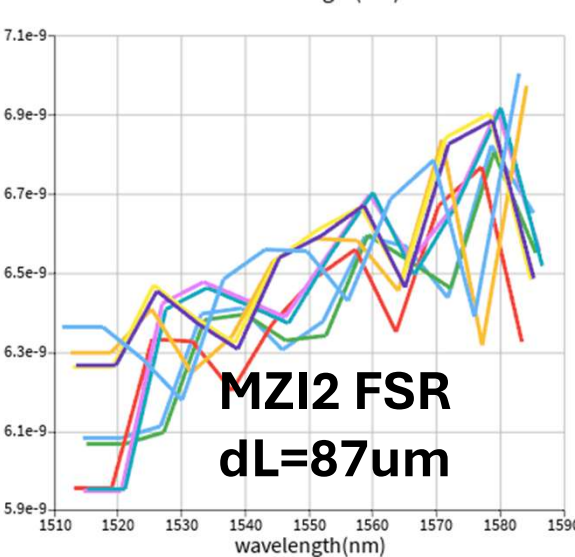
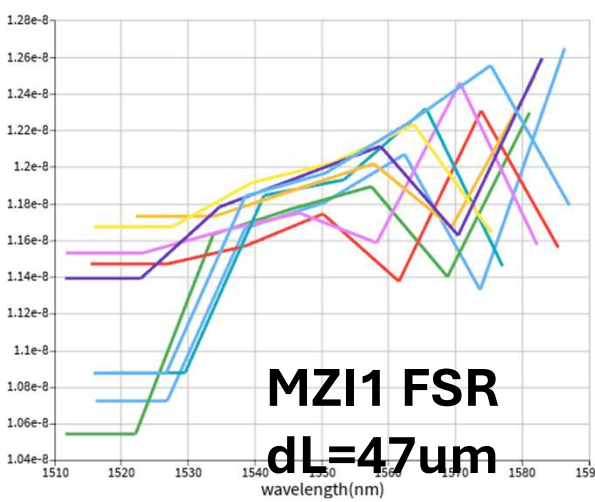
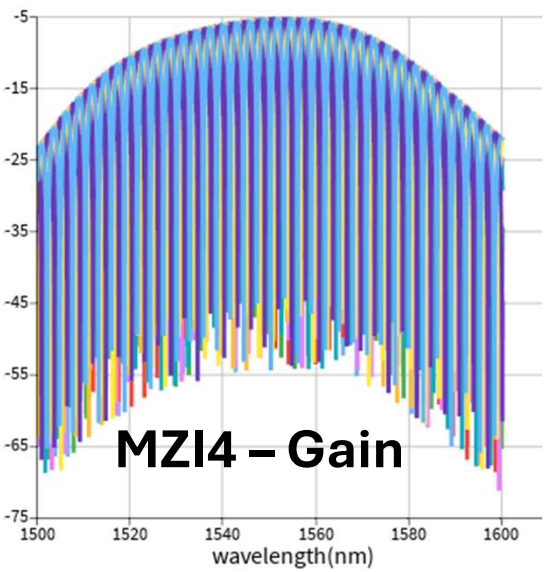
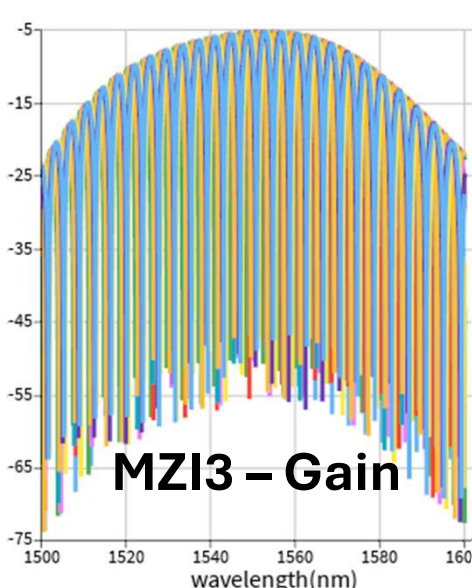
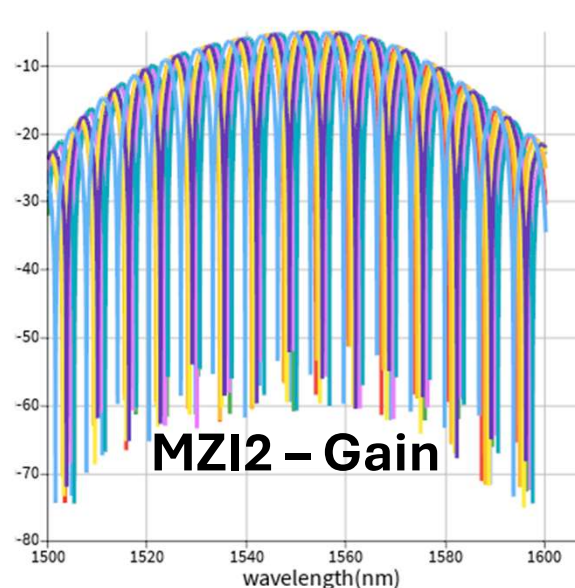
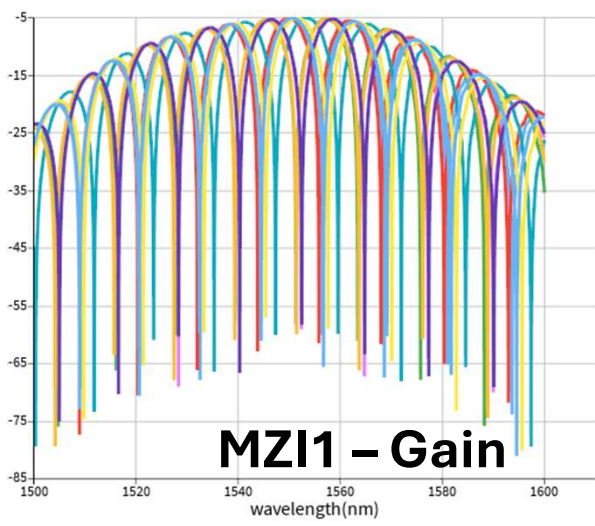


Lab results Vs. typical simulation fit. *Note: Lab Laser @ 1532nm, Sim @ 1550nm*



Device	L1[um]	L2[um]	FSR Sim [nm]	FSR0 Lab [nm]
MZI1	133.00	180.00	11.90	11.81
MZI2	133.00	220.00	6.57	6.44
MZI3	133.00	280.00	3.89	3.82
MZI4	133.00	330.00	2.89	2.85
MZI_SPLITTER	296.30	396.70	5.60	5.60

Corners simulations graphs: Gain , FSR



Group Index - Corners Simulations Vs. Lab Results $n_g(\text{typ})=4.2$

